FEU INSTITUTE OF TECHNOLOGY

COLLEGE OF COMPUTER STUDIES AND MULTIMEDIA ARTS

CS0065
(INTELLIGENT SYSTEMS)

TECHNICAL ASSESSMENT

3

CASE-BASED REASONING (CBR) SYSTEM DEVELOPMENT

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I. BACKGROUND INFORMATION

Case-Based Reasoning (CBR) is an intelligent problem-solving approach used in **Knowledge Representation and Reasoning (KRR)**. Instead of relying solely on predefined rules, CBR systems solve new problems by referring to past experiences, also known as cases. These cases contain previous problems and their successful solutions. When a new situation arises, the system identifies the most similar past case, adapts the old solution to fit the new context, and applies it. Afterward, the new case is saved, allowing the system to learn and continuously improve.

CBR is widely used in various fields, such as disaster response, legal assistance, and intelligent tutoring, because it mimics how humans naturally make decisions. For example, emergency responders rely on past typhoon data to plan evacuations; lawyers examine previous legal cases to build arguments; and tutoring systems look at prior student mistakes to provide customized guidance. In all these examples, past knowledge becomes the foundation for solving future problems more effectively.

This activity allows students to experience how CBR works by developing a simple, functional CBR program based on real-world scenarios. Through pair programming, students will learn how to represent cases, implement similarity assessment, adapt solutions, and store new learning experiences—reflecting the complete CBR cycle. This hands-on approach strengthens understanding of how intelligent systems use experiential knowledge to support reasoning, decision-making, and learning.

II. TECHNICAL ASSESSMENT

CASE-BASED REASONING (CBR) SYSTEM DEVELOPMENT

Objective

At the end of this laboratory activity, students are expected to:

1. **Understand and apply the Case-Based Reasoning (CBR) cycle**, including retrieval, reuse, revision, and retention.
2. **Represent cases effectively** using appropriate data structures such as dictionaries, arrays, JSON objects, or classes.
3. **Implement a similarity assessment method** to identify relevant past cases based on defined attributes.
4. **Adapt retrieved solutions** to meet the needs of a new problem scenario based on the chosen case study.
5. **Demonstrate learning in CBR** by saving the newly solved case into the system's case base.
6. **Develop collaboration skills** through pair programming and joint problem analysis.

7. **Apply knowledge representation and reasoning concepts** to real-world problems in disaster response, legal reasoning, or intelligent tutoring.

Materials Needed

1. Hardware
   - A laptop or desktop computer
   - Stable internet connection (for research, documentation, submission)

2. Software

A programming environment:
   - Python (recommended: IDLE, VS Code, Jupyter Notebook, PyCharm)
   - Any text editor (Notepad++, Sublime, VS Code) if needed

3. File Formats & Tools

Word processor for documentation:
   - MS Word, Google Docs, or LibreOffice
   - Screenshot tool (e.g., Snipping Tool, Lightshot, Print Screen)

4. Reference Materials

Lecture notes on:
   - Knowledge Representation & Reasoning
   - Case-Based Reasoning cycle
   - Similarity measures

Provided case studies (Disaster DSS, Legal DSS, Intelligent Tutoring System)

5. Pair Collaboration Tools (optional but useful)
   - Google Drive for shared documentation
   - GitHub/GitLab for code collaboration

General Instructions

1. This **Technical Assessment (TA3)** must be completed by Pair.
2. Please ensure that you are grouped as pairs in the People tab.
3. Select only one case study from the list below.
4. Include clear comments in your code to describe its logic.
5. Each program should display the correct input, process, and output.
6. Ensure that your script displays clear and readable outputs.
7. Compile all Python codes and corresponding outputs into one PDF file.
8. Save it as **TA3_lastname1_lastname2.pdf**
   (Example: TA2_Patilano_Loverez.pdf)
9. Submit the **python file** and **PDF** file.

📌 Step-by-Step Instructions

1. Form a pair

Choose a partner. Each pair will work collaboratively on research, coding, debugging, and presenting your results.

2. Choose ONE (1) case study

Select only one case study from the list below. Your program must be based on the scenario and logic of the chosen system:

Case Study Options:

1. *Disaster Response Decision Support System (DSS)*
2. *AI-Based Legal Decision Support System*
3. *Intelligent Tutoring System (ITS)*

3. Define your case base

Create at least **3–5 past cases** stored as dictionaries, JSON objects, arrays, or any structure.

Example structure:

```
{
  "problem": {...},
  "solution": {...}
}
```

4. Implement similarity assessment

Your program must include a way to measure similarity between the new case and past cases.

Examples:

- Attribute matching
- Weighted scoring
- Simple rule-based comparison

Output should show: (sample)

✔ Similarity score or ranking

✔ The most relevant past case retrieved

5. Adapt the solution

Modify the retrieved solution so it fits the new problem scenario.

This part should reflect the logic of your chosen case study:

Examples:

- **DSS:** Adjust evacuation plans based on population size
- **Legal DSS:** Modify arguments due to new digital evidence
- **ITS:** Give personalized feedback based on student errors

6. Demonstrate learning (retain stage)

After generating a final solution, store the new case into your case base.

Your program must show:

✔ "New case saved" or similar confirmation

✔ Updated case list

7. Display clear program outputs

Your output must include:

- New problem description
- Retrieved similar case
- Adapted final solution
- Updated case base

8. Submission Requirements

Submit the following:

A. Program File

- Python

B. Short Documentation (1–2 pages)

Include:

- Title of chosen case study
- Description of your case base
- Explanation of similarity method
- Adaptation logic
- Screenshots of program execution

C. Pair Reflection (5–7 sentences)

Discuss:

- What each partner contributed
- What challenges you encountered
- What you learned about CBR

Case Studies:

Case Study 1: Disaster Response Decision Support System

A CBR-based DSS helps local government units respond to natural disasters.

- Past cases include typhoons, floods, and earthquakes.
- New event: *Typhoon with expected flooding in urban areas.*
- System retrieves similar events: *Typhoon Ursula (2019)* and *Typhoon Odette (2021)*.
- It adapts recommended responses such as evacuation strategy, resource allocation, and shelter capacity.
- After execution, the system stores new outcomes for improved future reasoning.

Learning Outcome:

CBR helps governments respond faster by using *experiential knowledge* rather than rebuilding decisions from scratch.

Case Study 2: AI-Based Legal Decision Support System

A legal CBR tool assists lawyers by retrieving previous court cases similar to the current legal issue.

Scenario:

Client is charged with “theft involving electronic transfer.” System retrieves past cases involving:

- online theft
- digital fraud

- unauthorized bank transfers

Adaptation:
The reasoning engine adjusts argument patterns to match the new technical context.
Learning:
After final verdict, the case is stored—enhancing the system's knowledge for future cybercrime-related offenses.

Case Study 3: Intelligent Tutoring System
A CBR system identifies the learning struggles of students. Student submits a math solution with repeated errors. System retrieves past students with similar error patterns (e.g., misusing formulas). It adapts feedback and personalized lessons based on successful interventions. **Learning:** System improves by retaining new student profiles and outcomes.

Your Code - Laboratory Tasks ⇩

```
"""
CS0065 - TA3: Case-Based Reasoning (CBR)
Case Study 1: Disaster Response Decision Support System (simple version)

CBR cycle: Retrieve -> Reuse (adapt) -> Retain
Case values are simplified, illustrative figures, not official statistics.
"""

import math

case_base = [
    {
        "name": "Typhoon Ursula (2019)",
        "problem": {
            "type": "typhoon",
            "area": "urban",
            "population": 120000
            },
        "solution": {
                "strategy": "Evacuate low-lying barangays to schools and
gyms",
            "evacuee_percent": 0.30,
            "shelter_capacity": 400
```

```
            },
    },
    {
        "name": "Typhoon Odette (2021)",
        "problem": {
            "type": "typhoon",
            "area": "coastal",
            "population": 250000
            },
        "solution": {
            "strategy": "Move coastal residents to inland shelters",
            "evacuee_percent": 0.45,
            "shelter_capacity": 350
            },
    },
    {
        "name": "Metro Flood (2012)",
        "problem": {
            "type": "flood",
            "area": "urban",
            "population": 300000
            },
        "solution": {
            "strategy": "Evacuate riverside communities and use boats",
            "evacuee_percent": 0.25,
            "shelter_capacity": 450
            },
    },
    {
        "name": "Bohol Earthquake (2013)",
        "problem": {
            "type": "earthquake",
            "area": "rural",
            "population": 80000
            },
        "solution": {
            "strategy": "Move residents to open ground and set up tents",
```

```
            "evacuee_percent": 0.40,
            "shelter_capacity": 250
            },
    },
]

# ------------------------------------------------------------
def similarity(new, old):
    score = 0.0
    # Same disaster type is the most important (weight 0.5)
    if new["type"] == old["type"]:
        score += 0.5

    # Same area type (weight 0.3)
    if new["area"] == old["area"]:
        score += 0.3

         score  +=  0.2  *  min(new["population"],  old["population"])  /
max(new["population"], old["population"])
    return score

def adapt(new, old_case):
    old = old_case["solution"]
    evacuees = round(new["population"] * old["evacuee_percent"])
    shelters = math.ceil(evacuees / old["shelter_capacity"])
    strategy = old["strategy"]

    # Urban flooding needs extra measures
    if new["area"] == "urban":
        strategy += "; add vertical evacuation and rescue boats"
                return    {"strategy":    strategy,    "evacuee_percent":
old["evacuee_percent"],
            "shelter_capacity": old["shelter_capacity"],
            "evacuees": evacuees, "shelters_needed": shelters}

# ------------------------------------------------------------
def main():
```

```
    new_problem = {
        "type": "typhoon",
        "area": "urban",
        "population": 200000
        }

    print("=== CBR DISASTER RESPONSE SYSTEM ===")
      print("\nNEW PROBLEM: Typhoon with expected flooding in an urban
area")
    print(f"  Type: {new_problem['type']}, Area: {new_problem['area']}, "
          f"Population: {new_problem['population']:,}")

    # RETRIEVE
    print("\nSIMILARITY SCORES:")
       ranked = sorted(case_base, key=lambda c: similarity(new_problem,
c["problem"]), reverse=True)
    for i, c in enumerate(ranked, 1):
              print(f"    {i}.  {c['name']:<26}  {similarity(new_problem,
c['problem']):.2f}")
    best = ranked[0]
    print(f"\nMOST SIMILAR CASE: {best['name']}")
    print(f"  Old strategy: {best['solution']['strategy']}")

    # REUSE
    final = adapt(new_problem, best)
    print("\nADAPTED FINAL SOLUTION:")
    print(f"  Strategy : {final['strategy']}")
                print(f"         Evacuees     :     {final['evacuees']:,}
({final['evacuee_percent']:.0%} of population)")
         print(f"     Shelters   :   {final['shelters_needed']}   (capacity
{final['shelter_capacity']} each)")

    # RETAIN
    case_base.append({"name": "Typhoon New Event (2025)",
                      "problem": new_problem, "solution": final})
    print("\nNew case saved!")
    print("\nUPDATED CASE BASE:")
```

```
    for i, c in enumerate(case_base, 1):
        print(f"  {i}. {c['name']}")

if __name__ == "__main__":
    main()
```

B. Documentation:

Title of Chosen Case Study

Case Study 1: Disaster Response Decision Support System (DSS)

Description of the Case Base

The case base contains four past disasters: Typhoon Ursula (2019), Typhoon Odette (2021), Metro Flood (2012), and the Bohol Earthquake (2013). Each case is stored as a Python dictionary with a name, a problem, and a solution.

The problem part describes the event using three attributes, which are the disaster type, the area type (urban, coastal, or rural), and the population. The solution part records the response that was used, including the evacuation strategy, the percentage of the population that was evacuated, and the capacity of each shelter.

The values are simplified estimates made for this exercise and are not official statistics.

Similarity Method

The system uses weighted attribute matching to find the past case that is closest to the new problem. Each attribute adds points to a score between 0 and 1. Disaster type is worth 0.5 and area type is worth 0.3, and both are all-or-nothing matches.

Disaster type has the highest weight because the kind of hazard changes the whole response, and area type comes next because urban, coastal, and rural areas need different tactics. Population is worth up to 0.2 and is computed by dividing the smaller population by the larger one, so closer population sizes score higher.

For the new problem, a typhoon in an urban area with 200,000 residents, Typhoon Ursula scored 0.92, Typhoon Odette 0.66, Metro Flood 0.43, and the Bohol Earthquake 0.08. Ursula had the highest score, so it was retrieved as the most similar case.

Adaptation Logic

After retrieving Typhoon Ursula, the system adapts its solution to fit the new event. It reuses Ursula's evacuation rate of 30% and applies it to the new population, which gives 60,000 evacuees (200,000 × 0.30). It then divides the number of evacuees by the shelter capacity of 400 and rounds up, giving 150 shelters needed. Because the new area is urban, the system also adds vertical evacuation and rescue boats to the strategy. Finally, the adapted solution is saved as a new case, so the case base grows from four cases to five and can be used for future problems.

C. Pair Reflection (5–7 sentences)

Each partner contributed to this activity equally, Mr. Hizon mostly focused on the code while Mr. Coritana focused more on answering the PDF file like the documentation and reflection. But both partners still gave their own opinions and takeaways on the other partner's part. The most challenging part of this activity was about the actual algorithm to determine which scenarios would be used as a reference for the CBR system — like getting the calculations according to the weights of each past problem.

What we learned about CBR is that this is a really useful strategy that can be implemented in the real world that can positively affect millions of lives. If scenarios and problems occur, if we have previous data as well as CBR implemented, these problems would be easily solved. As they say, experience is the best teacher — we agree and the data shows!

Screenshot

```
=== CBR DISASTER RESPONSE SYSTEM ===

NEW PROBLEM: Typhoon with expected flooding in an urban area
  Type: typhoon, Area: urban, Population: 200,000

SIMILARITY SCORES:
  1. Typhoon Ursula (2019)      0.92
  2. Typhoon Odette (2021)      0.66
  3. Metro Flood (2012)         0.43
  4. Bohol Earthquake (2013)    0.08

MOST SIMILAR CASE: Typhoon Ursula (2019)
  Old strategy: Evacuate low-lying barangays to schools and gyms

ADAPTED FINAL SOLUTION:
  Strategy : Evacuate low-lying barangays to schools and gyms; add vertical evacuation and rescue boats
  Evacuees : 60,000 (30% of population)
  Shelters : 150 (capacity 400 each)

New case saved!

UPDATED CASE BASE:
  1. Typhoon Ursula (2019)
  2. Typhoon Odette (2021)
  3. Metro Flood (2012)
  4. Bohol Earthquake (2013)
  5. Typhoon New Event (2025)
```